

A rational scalar determinant is not integral

Disproof of conjecture 00000001645

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Take the trivial group $G = \{e\}$ and the rational group-ring matrix $A = [\frac{1}{2}e] \in M_1(\mathbb{Q}G)$. The group is torsion-free. Moreover, $\ell^2(G) = \mathbb{C}$, its group von Neumann algebra is $\mathbb{C}$, and its canonical trace is $\tau(z) = z$. For arbitrary rectangular matrices over this algebra the von Neumann dimension of the kernel is its ordinary complex dimension: the extended canonical trace is the usual, unnormalized matrix trace, and the trace of an orthogonal projection is its rank. Thus G satisfies the strong Atiyah conjecture, including its integer-dimension requirement for torsion-free groups.

The regular representation of $\frac{1}{2}e$ is genuinely multiplication by $\frac{1}{2}$: convolution gives

$$(\lambda(\tfrac{1}{2}e)f)(x) = \sum_{g \in G} (\tfrac{1}{2}e)(g)f(g^{-1}x) = \tfrac{1}{2}f(x).$$

In the orthonormal basis δ_e , this operator is the matrix $T = [\frac{1}{2}]$. It is invertible, $T^*T = [\frac{1}{4}]$, and therefore $|T| = (T^*T)^{1/2} = [\frac{1}{2}]$. The defining spectral-calculus formula for the Fuglede–Kadison determinant consequently gives

$$\det_{\tau}^{\text{FK}}(T) = \exp(\tau(\log |T|)) = \exp(\log(\tfrac{1}{2})) = \tfrac{1}{2}.$$

This uses the canonical normalized trace $\tau(1) = 1$; there is no matrix normalization ambiguity in size one.

A rational number is an algebraic integer exactly when it is an integer. Hence $\frac{1}{2}$ is not an algebraic integer, and in particular is not an algebraic unit. This contradicts the conjecture’s explicit assertion about *all rational-coefficient matrices*. Even if the preceding integrality clause were intended only for integral group-ring matrices, the rational algebraic-unit clause would still be false. The usual determinant conjecture for integral group rings is a lower-bound statement, not the rational claim made here; no such lower-bound statement is being refuted.

Formal certificate. The Lean project constructs the trivial group, its actual rational monoid algebra, convolution and regular matrix, canonical trace and the singleton spectral-calculus logarithm. It proves the determinant value and its failure to be integral over $\mathbb{Z}$, using integrally closedness of $\mathbb{Z}$ in $\mathbb{Q}$ and the injective embedding into $\mathbb{R}$. It also records integer kernel dimensions for every finite rectangular complex matrix, the finite-dimensional interpretation of Atiyah’s requirement above.

Reference. P. de la Harpe, *The Fuglede–Kadison determinant*, arXiv:1107.1059, Section 3, formula (6) and its scalar consequence.